

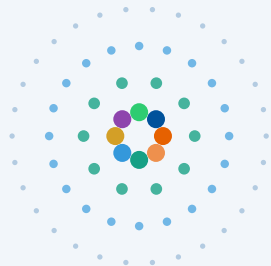
From Episodes to Abstractions

Latent Hierarchical Memory in 1,908 AI Conversations

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- outer: 1,908 episodes
- 500 meta-concepts
- 50 themes
- inner: 8 domains

 github.com/queelius/chatgpt-complex-net

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The Problem: Archives Are Flat. Knowledge Is Not.

What you have

Chronological list

Bayesian inference

Cooking pasta

Solomonoff induction

R package debugging

Philosophy of mind

⋮

Structure is invisible.

extract
→
concepts

The Question

Does **structure** emerge from a long archive of AI conversations — the way it does in human memory?

What's hidden

Latent hierarchy



If it's there, we should find it.

Two Ideas from Cognitive Science

Hippocampal

fast, episodic
specific events

consolidate



Neocortical

slow, statistical
extracts regularities

Prediction:

New conversations reuse *existing* concepts
more and more — vocabulary growth **slows**

[McClelland et al. 1995; Kumaran et al. 2016]

**And: human semantic memory is
a small-world network**



Dense local clusters + a few global shortcuts.

Roget's, WordNet, word associations:

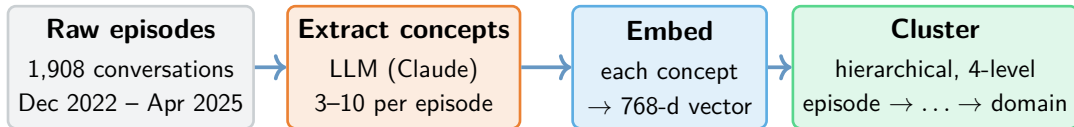
small-worldness $\approx 5-15$

[Steyvers & Tenenbaum 2005]

The Hypothesis

If AI conversation archives are an *externalised* semantic memory, we should see both
signatures: **sublinear vocabulary growth** and **small-world topology**.

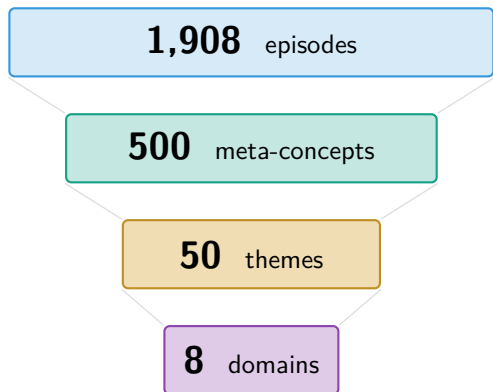
Pipeline: Concepts $\rightarrow$ Embeddings $\rightarrow$ Hierarchy



Key instruction in the prompt: *“Be specific. Say ‘Metropolis–Hastings algorithm,’ not ‘statistics.’”*

Output: a graph linking each episode to its concepts,
then grouped into a four-level hierarchy.

The Reveal: A Four-Level Hierarchy Emerges



*Ward linkage on concept embeddings,
cut at 3 heights · $\sim 3.5\times$ compression per level*

Eight interpretable domains

Software Engineering	1,074
Math & Optimisation	634
Statistical Methods	510
Philosophy & AI Theory	489
ML & Network Science	274
DevOps & Integration	273
AI Safety & Research	177
LLM Engineering	86

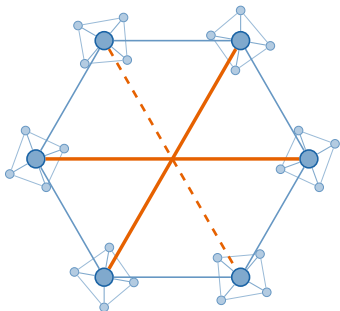
Two surprises

- Domains are **easy to name** — but there's no clean number of clusters.
- Most episodes **belong to several domains at once**.

Signature 1: Small-World Topology

Concept co-occurrence network

499 meta-concept nodes · 5,935 co-occurrence edges



Node: 1 of 500 meta-concepts

Edge: both concepts appear in the same episode

Compared to a random graph

[Humphries & Gurney 2008]

Local clustering	6.89×
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Shortest paths	1.05×
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Small-worldness	6.6
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Matches human memory

(small-worldness)

Roget's Thesaurus ≈ 13

WordNet ≈ 15

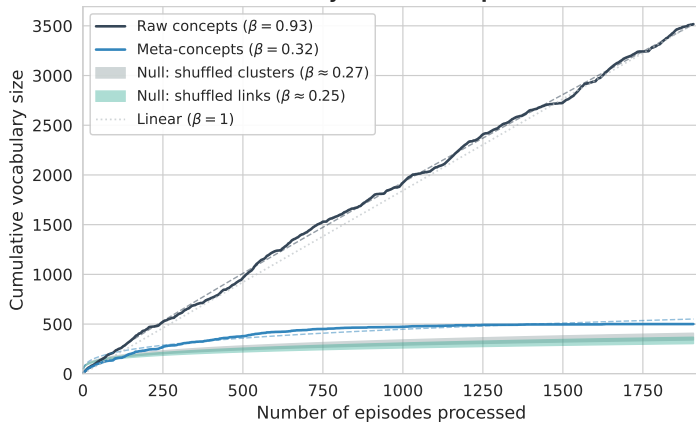
Word associations ≈ 5.6

Our archive ≈ 6.6

[comparison values: Steyvers & Tenenbaum 2005]

Signature 2: Sublinear Vocabulary Growth

Vocabulary Growth: Heaps' Law



Heaps' law: $V(n) = K \cdot n^\beta$

[Heaps 1978]

Raw concepts $\beta = 0.93$

Meta-concepts $\beta = 0.32$

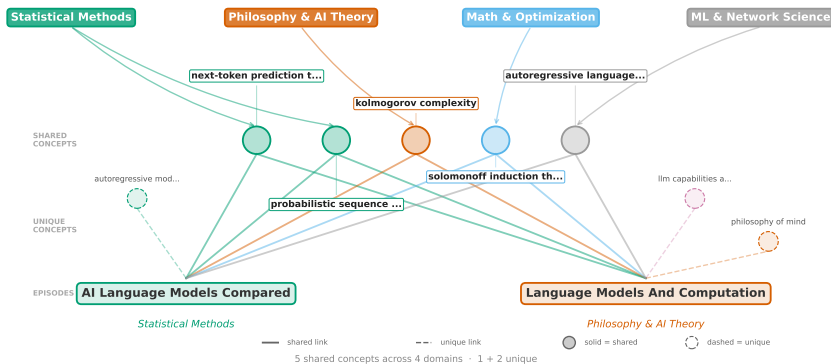
Caveat: with k fixed at 500

Some saturation is automatic. So we compare against **null models**.

Two nulls, both reject

- Shuffled clusters: $\beta = 0.27$
- Shuffled links: $\beta = 0.25$
- Both $p < 10^{-3}$

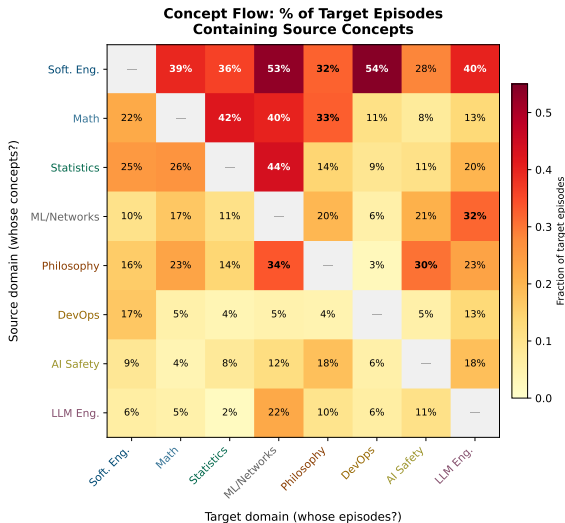
77% of Episodes Span Multiple Domains



One-label methods (*Louvain, k-means, BERTopic*): each episode gets one cluster. These two end up apart — connection vanishes.

Set-based methods: each episode is a set of concepts. Two episodes share concepts across four domains and stay linked.

Who Borrows From Whom? (Concept Flow)



Two questions per domain

Where do its concepts go? (rows)

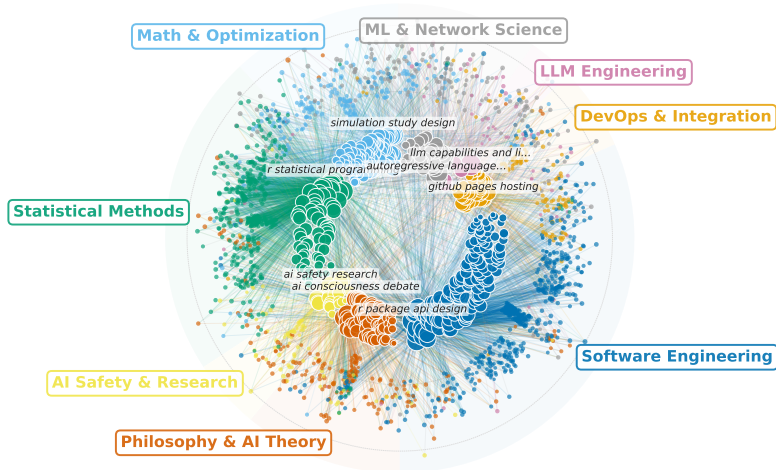
Which concepts come in? (columns)

What the heatmap says

- On average **39% below** random
⇒ domains really are separate
- LLM Eng. → ML/Net: **2.67×** random
⇒ specific dependencies

Software Eng. broadcasts to all 7 others.

The Full Picture



Inner: 499 concepts (coloured by domain). Outer: 1,908 episodes. Edges: episode→concept.

Takeaways & What's Next

What we found

- 4-level hierarchy
1,908 → 500 → 50 → 8
- Small-world structure matches human memory
- Vocabulary saturates as CLS predicts
- 77% span ≥ 2 domains

Why it matters

- Archives = **externalised semantic memory**
- **Concept-level retrieval** (Graph RAG)
- Knowledge is a **continuum, not a partition**

Limits & next

- **N=1** single user.
Replicate on WildChat
- **Cluster count** is a design choice

Treat AI conversation archives as **structured cognitive artifacts** — the structure was there all along; we just needed the right lens.

Thank you